

Fraud Alert Vulnerability Index

Analytical Progress and Feature Engineering Report

Reporte de Progreso Analitico y Feature Engineering

Confluence-ready bilingual report

This document consolidates the tenure analysis, trades structural EDA, MOP behavioral analysis, rolling-window feature development, key decisions, limitations, and the next steps required to build the alert-propensity index. / Este documento consolida el analisis de tenure, EDA estructural de trades, comportamiento MOP, ventanas temporales, decisiones, limitaciones y los proximos pasos para construir el indice de propension a alerta.

Prepared from Spark/Hive analyses and business rules supplied during the project.

25-Aug-2026

Version en Espanol

Propiedad	Valor
Estado	En desarrollo - EDA y feature engineering
Objetivo	Construir un indice/modelo point-in-time de propension a colocar una primera alerta de fraude.
Fuentes principales	security_alert, person_restricted, trades
Periodo analizado	Historia fuente 2000-2026; EDA conductual detallado 2024-2025
Fecha de actualizacion	24-Aug-2026

1. Resumen ejecutivo

- La fuente de alertas es un historico de snapshots; por ello se deduplicaron eventos de creacion y se uso la primera alerta observada por party_id como outcome inicial.
- El analisis de tenure encontro una censura izquierda material en person_restricted: 29,086 de 66,643 parties de la muestra (43.64%) aparecen exactamente en la primera fecha historica disponible, 2000-09-30.
- Al excluir censura y casos inconsistentes quedaron 37,492 parties con tenure utilizable. La mediana de tenure al primer alert fue 195 meses (~16.3 anos), pero el pico se desplaza +12 meses por cada ano calendario, indicando un fuerte efecto de cohorte y no un umbral causal de tenure.
- trades tiene una carga mensual y permite construir comportamiento 3m/6m/12m. data_date se interpreta como fecha de disponibilidad/carga y eff_dte como fecha efectiva de ultima verificacion de la cuenta.
- La frecuencia de actualizacion efectiva esta fuertemente explicada por el ciclo de vida de la cuenta: 96.61% de las cuentas STATIC estan cerradas; entre cuentas sin fecha de cierre, 86.06% son MONTHLY.
- Las transiciones de MOP son una senal conductual limpia: 99.98% de los deterioros coinciden con una nueva eff_dte. A nivel persona, 15.15% de la muestra 2025 experimento al menos un deterioro.
- La capa 2024-2025 tiene 707,157 party-months, 31,189 parties, 24 meses y cero duplicados por party_id + data_date. El 96.58% de los party-months 2025 tiene un baseline completo de 12 meses.
- Entre observaciones con baseline completo, 8.09% tiene deterioro reciente, 6.12% muestra aceleracion del deterioro frente a su propio baseline y 2.10% presenta un evento adverso reciente.

Decision metodologica principal

Tenure debe mantenerse como variable de contexto/segmentacion. Las candidatas principales para el indice son cambios conductuales recientes respecto al baseline individual: deterioro MOP, past due, balance/utilizacion, eventos adversos y cambios en el portafolio.

2. Fuente de alertas y definicion del outcome

Metrica	Resultado
Filas source security_alert	324,268,295
Eventos unicos de creacion	42,318,245
Parties con alerta	29,766,492
Min / max crat_dte	2007-01-13 / 2026-07-30
Min / max data_date	2009-03-31 / 2026-07-31
Cohorte primera alerta 2022-2025	6,663,521 parties
Tipo 2 en cohorte	6,538,126 (98.12%)
Tipo 3 en cohorte	122,420 (1.84%)
Tipos 2 y 3 mismo dia	2,975 (0.04%)

- La regla de negocio indica que, despues de colocar una alerta, una persona no puede colocar otra durante aproximadamente 7 anos.
- Se encontraron muchas fechas de creacion dentro de 7 anos, por lo que una nueva crat_dte no debe interpretarse automaticamente como una nueva alerta independiente. Para el objetivo actual se utiliza la fecha mas temprana como primera alerta observada.
- El outcome actual representa propension a colocar una primera alerta observada; no equivale por si solo a fraude confirmado o victimizacion confirmada.

3. Tenure: resultados y cambio de interpretacion

Indicador	Resultado
Muestra deterministica de alertas	66,643 parties

Left-censored exacto en 2000-09-30	29,086 (43.64%)
No censurados	37,557
Validos despues de quality filters	37,492
Promedio tenure	175.23 meses (~14.6 anos)
P10 / P25	61 / 121 meses
P50	195 meses (~16.3 anos)
P75 / P90 / P95	230 / 257 / 270 meses

Ano de alerta	Banda pico de tenure	% dentro del ano
2022	180-191 meses	12.39%
2023	192-203 meses	11.93%
2024	204-215 meses	11.74%
2025	216-227 meses	10.20%

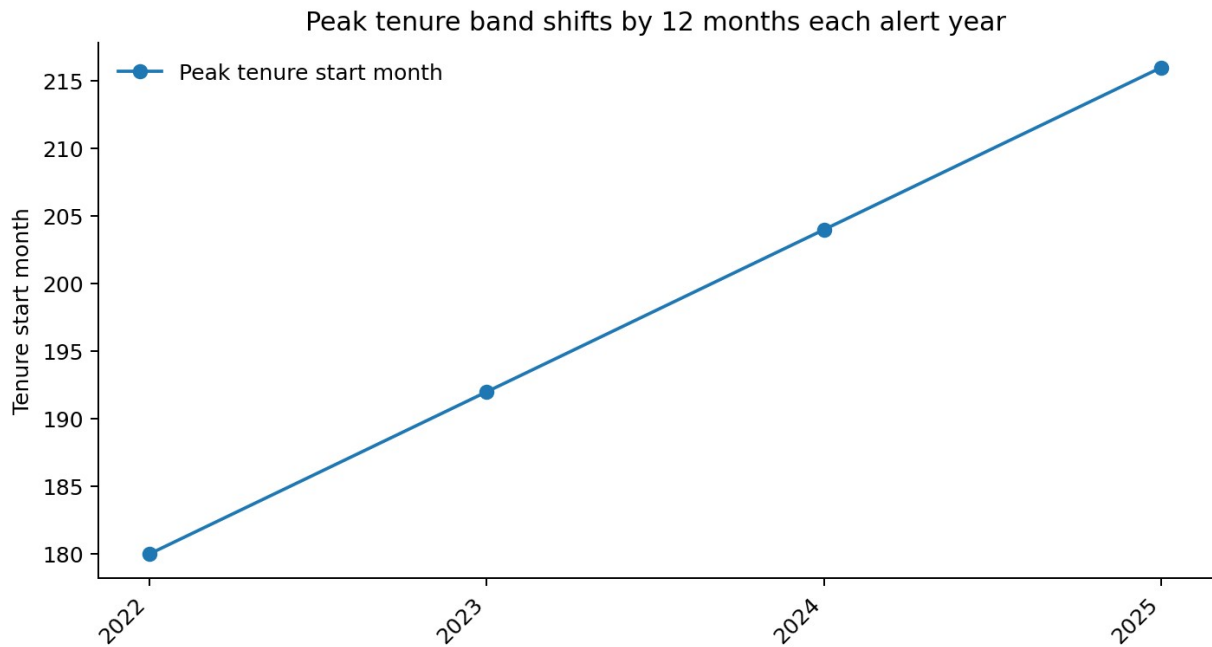


Figura 1. El pico de tenure avanza exactamente 12 meses por ano, consistente con un efecto de cohorte.

Interpretacion

No se debe concluir que 17-19 anos es el periodo de mayor riesgo. El desplazamiento anual indica que una cohorte numerosa (aprox. originada 2006-2007) esta envejeciendo con el calendario. Para medir riesgo por tenure se necesita el denominador de personas expuestas, incluyendo quienes no colocan alerta.

4. trades: EDA estructural y temporal

Metrica	Resultado
Filas trades	504,842,538,762
Frecuencia de carga	Mensual
Historia observada	Desde 2000-09 hasta 2026
missing eff_dte	0
eff_dte > data_date	20 filas; practicamente 0%
Llave validada en muestra	party_id + acct_id + data_date; 0 duplicados

Percentil	DATEDIFF(data_date, eff_dte)
P10	12 dias
P25	25 dias
P50	275 dias
P75	1,671 dias
P90	2,615 dias
P95	3,073 dias
P99	3,501 dias

Lectura correcta de eff_dte

La diferencia data_date - eff_dte se interpreta como staleness/frescura de la informacion, no como simple load lag. Una carga mensual puede repetir el mismo estado efectivo durante meses o años.

5. Ciclo de vida de cuenta y frecuencia de actualizacion

Update behavior	Cuentas	%
STATIC (1 eff_dte)	167,962	51.50%
INTERMEDIATE (2-9)	29,323	8.99%
MONTHLY (10-12)	128,834	39.51%

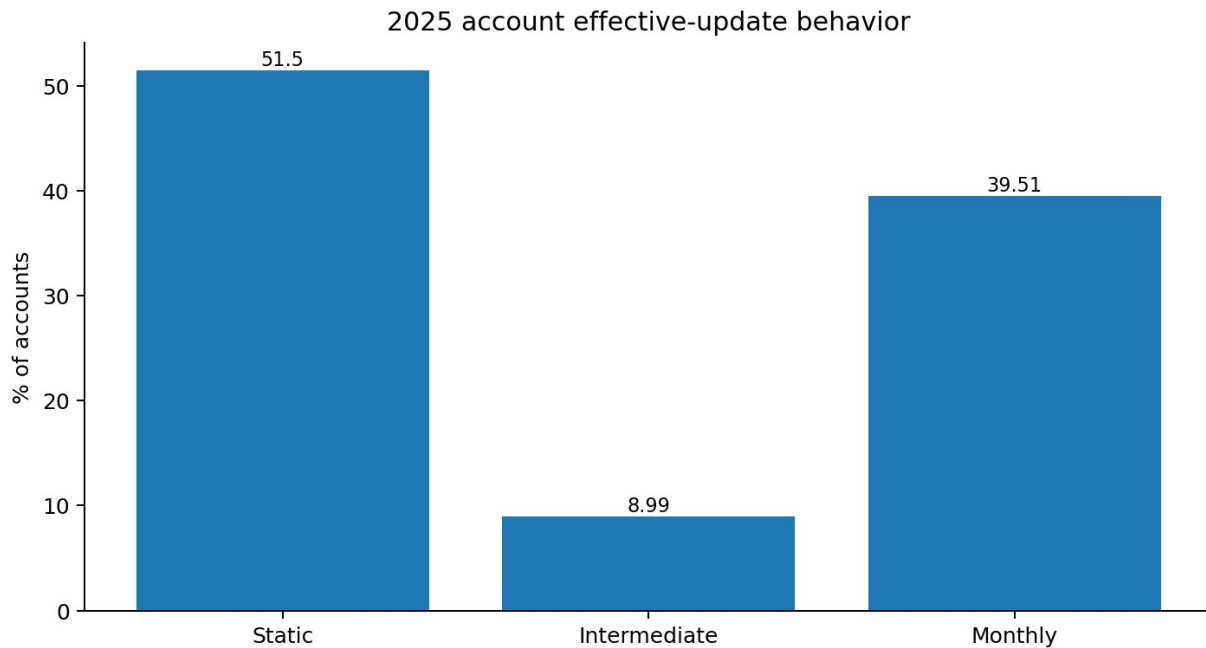


Figura 2. Distribucion de frecuencia efectiva de actualizacion en la muestra 2025.

Grupo	Closed	Zero balance	Past due
STATIC	96.61%	99.06%	0.69%
INTERMEDIATE	55.87%	61.38%	7.79%
MONTHLY	10.76%	26.31%	8.63%

Grupo (solo sin clsd_dte)	Cuentas	%
STATIC	5,689	4.26%
INTERMEDIATE	12,940	9.69%
MONTHLY	114,968	86.06%

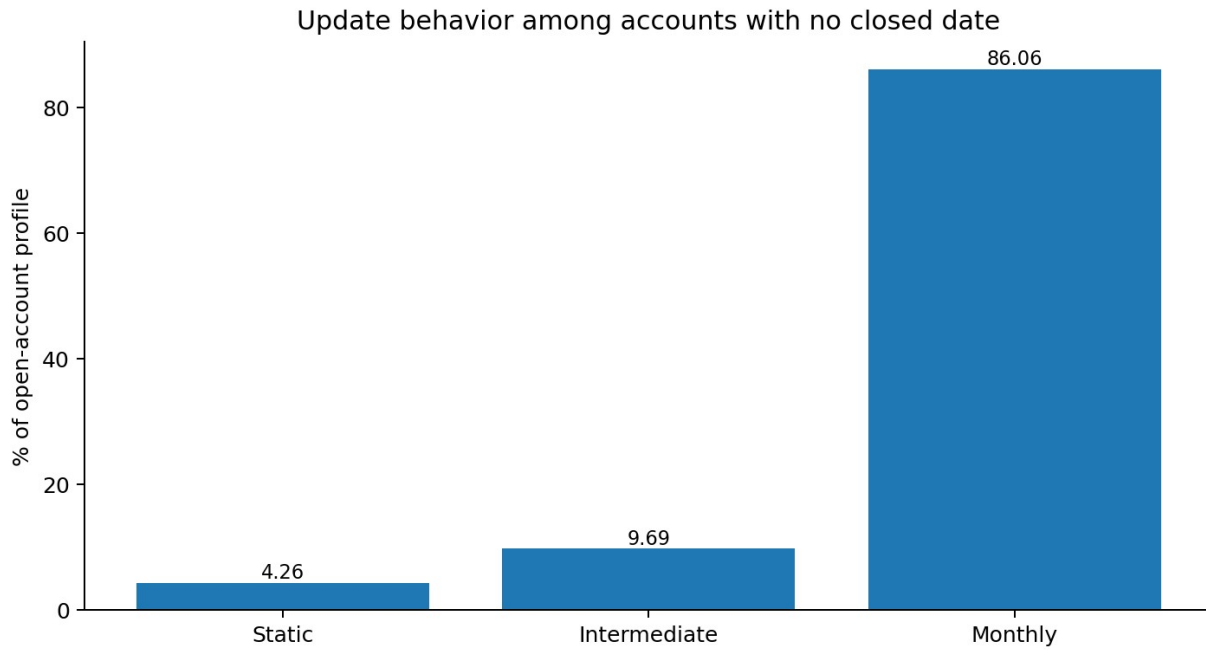


Figura 3. Al controlar por cuentas sin fecha de cierre, la actualizacion mensual es el comportamiento dominante.

Conclusion

update_behavior es principalmente una variable de ciclo de vida/reporting, no una senal directa de vulnerabilidad. Se conserva como contexto; la senal de mayor valor sera el cambio respecto al patron historico de la misma cuenta/persona.

6. Primer EDA conductual: transiciones de MOP

Tipo de transicion	Transition count	%
NO CHANGE	3,501,681	98.79%
DETERIORATION	29,310	0.83%
IMPROVEMENT	7,843	0.22%
NEW_ADVERSE_STATUS (v1)	4,042	0.11%
OTHER_TRANSITION	1,582	0.04%

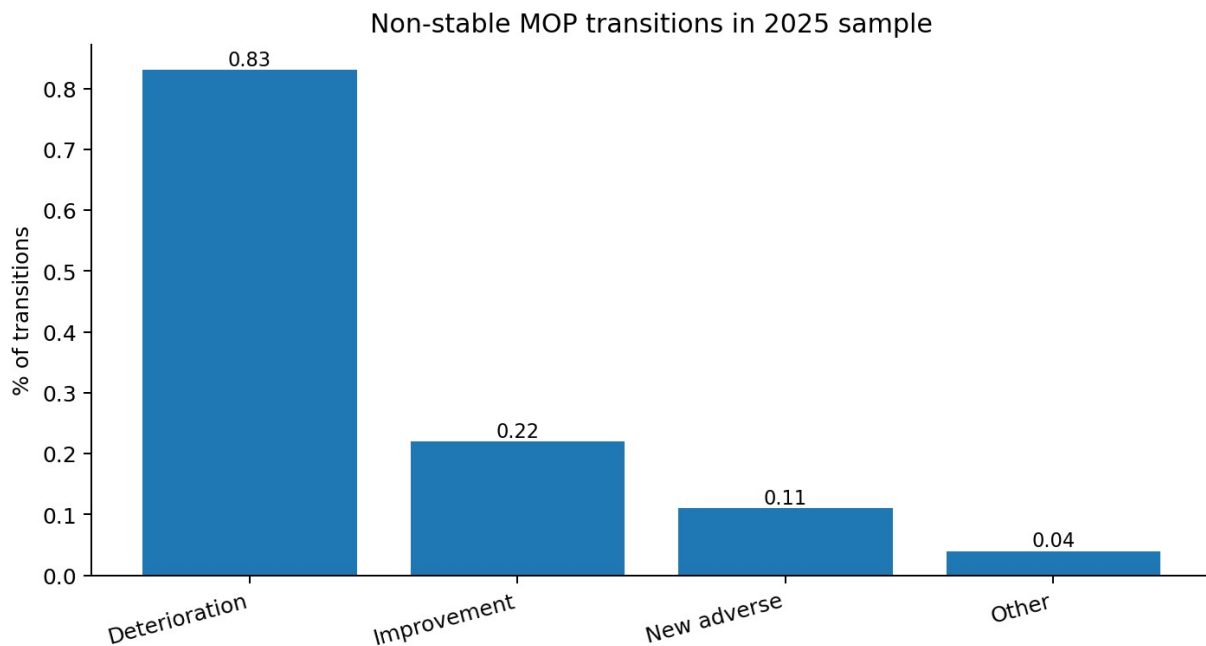


Figura 4. Las transiciones no estables son raras a nivel cuenta-mes, pero informativas.

Validacion temporal	% con effective_update_flag=1
Deterioration	99.98%
Improvement	99.78%
New adverse status	99.01%

- La logica ordinal 01-06 se valido para deterioro/mejora. Los cambios observados mas comunes incluyen 01->02, 04->05, 02->03, 01->04 y 03->04.
- Se corrigio la definicion de adverse: 09, 9B, 8A y 8B representan estados adversos activos; 9P y 8P se separan como historico/resuelto.
- Los codigos 07 y 08 permanecen sin interpretar hasta contar con una definicion oficial.

Evento MOP corregido	Count
DPD deterioration	29,310
DPD improvement	7,843
Entered active adverse	3,394
Adverse resolved	477
Became unrated	1,136
Left unrated	107

7. Agregacion a persona-mes y extension 2024-2025

Metrica	Resultado
Parties 2025 analizados en EDA MOP	30,496
Parties con al menos un deterioro	4,624 (15.15%)
Parties con active adverse	1,697 (~5.6%)
P90 / P95 / P99 deteriorated accounts por party	2 / 6 / 19

Capa mensual 2024-2025	Resultado
party-month rows	707,157
parties	31,189
meses	24
rango	2024-01-31 a 2025-12-31
duplicados party_id + data_date	0

Eventos 2025 con historia 2024 disponible	Count
Deteriorations	31,437
Improvements	8,331
Active adverse	3,893

Boundary effect corregido

Al incorporar 2024, enero de 2025 ya puede compararse con diciembre de 2024. El incremento de eventos 2025 es esperado y refleja mejor cobertura longitudinal, no una inconsistencia.

8. Rolling windows y baseline conductual

Cobertura 2025	Party-months	%
Baseline 12m completo	343,948	96.58%
Incompleto	12,173	3.42%

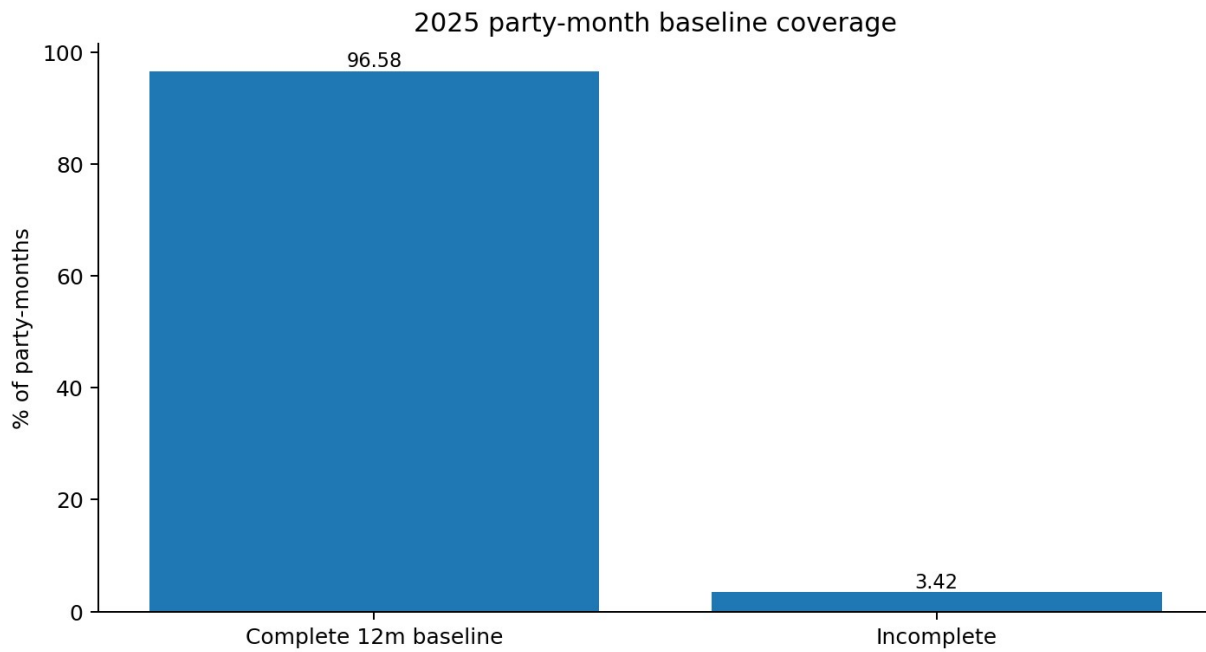


Figura 5. La cobertura de historia completa es suficiente para comparar ventanas recientes contra baseline.

Senal conductual	Party-months	%
Deterioro en recent 3m	27,838	8.09%
Aceleracion de deterioro > 0	21,053	6.12%
Active adverse en recent 3m	7,216	2.10%

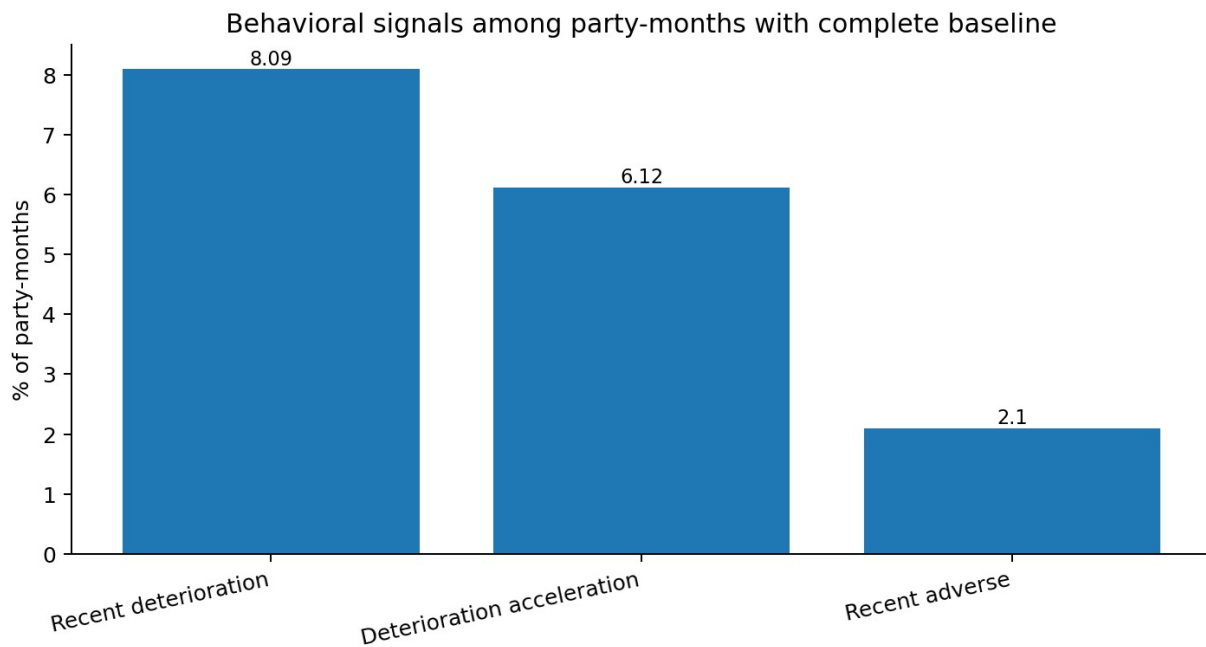


Figura 6. Frecuencia de las primeras features conductuales con baseline completo.

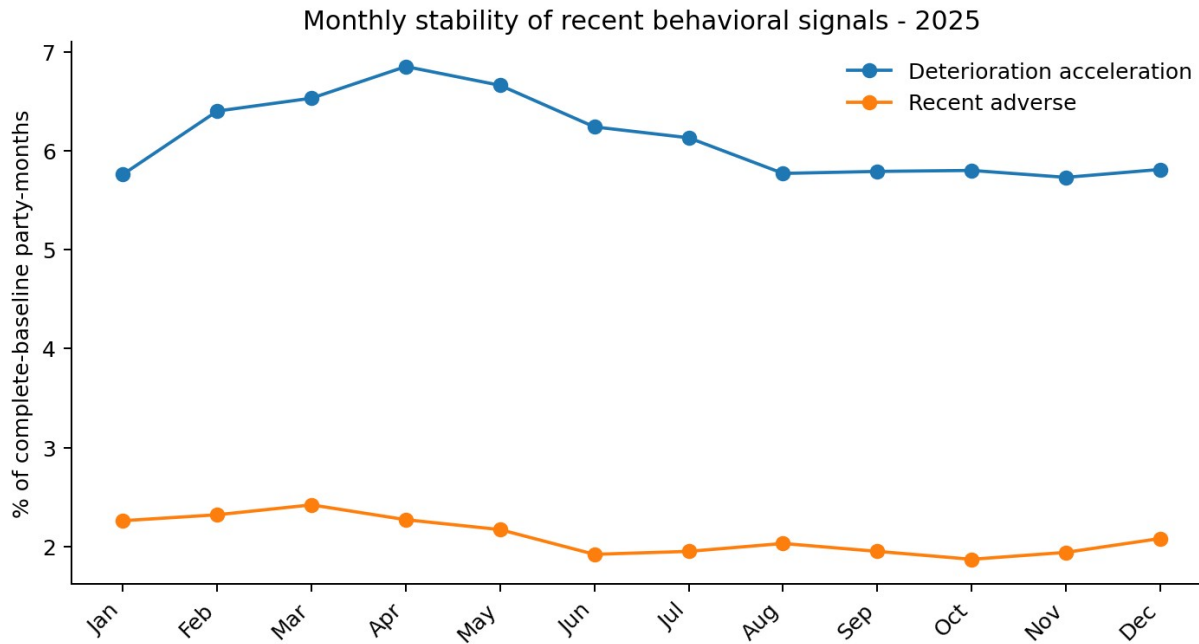


Figura 7. La aceleracion de deterioro y los adverse recientes muestran estabilidad mensual razonable durante 2025.

- pct_deterioration_acceleration oscila aproximadamente entre 5.73% y 6.85% por mes en 2025.
- pct_recent_adverse oscila aproximadamente entre 1.87% y 2.42%.
- P95 de deterioration_acceleration permanece alrededor de 0.2-0.3 durante el año, lo que sugiere una feature relativamente estable a nivel calendario.
- El output pegado de percentiles de magnitud para past_due/balance/utilization quedo mal formateado; se excluye deliberadamente del reporte hasta re-exportarlo de forma limpia.

9. Feature Catalog - estado actual

Familia / feature	Prioridad	Decision actual
tenure_months	Media	Contexto/segmentacion; no usar como riesgo aislado
active_accounts	Alta	Control de exposicion y denominador
deteriorations_3m/6m/12m	Muy alta	Candidata principal
deterioration_acceleration	Muy alta	Cambio recent 3m vs prior 9m
max_dpd_jump_3m	Alta	Severidad del deterioro observado
entered_active_adverse_3m/6m	Muy alta	Evento severo reciente
past_due level/change	Muy alta	Nivel y cambio vs baseline
revolving utilization/change	Muy alta	Solo portfolios R/C; validar outliers
new / recently closed accounts	Alta	Pendiente de feature engineering
update_behavior raw	Baja	Lifecycle/reporting, no riesgo directo
staleness / update frequency change	Experimental	Usar solo como cambio relativo al baseline
legis_prhib / medical classifications	Excluir	Gobernanza/sensibilidad; no score
DOB/age/state	Auditoria	No incluir por defecto en primera version productiva

10. Proximo paso para construir el indice

- 1) Re-exportar y validar percentiles de past_due_change, balance_change, utilization_change y delinquency_share_change.
- 2) Crear la tabla point-in-time de observacion. Para cada alerta, usar el ultimo data_date estrictamente anterior a first_alert_date; nunca usar una carga posterior al evento.
- 3) Construir labels alert_next_1m, alert_next_3m, alert_next_6m y alert_next_12m. 1 = primera alerta dentro de la ventana; 0 = sin alerta con follow-up completo; NULL = censurado.
- 4) Construir el universo de controles/no-alerta y el denominador de elegibilidad. Esto permite pasar de concentracion descriptiva a riesgo/tasa real.
- 5) Llevar las features 3m/6m/12m y recent-3m-vs-prior-9m a la fecha de observacion, garantizando point-in-time y evitando leakage.

- 6) EDA por label: distribuciones, lift, event rate por bandas, missingness, monotonicidad, estabilidad por mes/año y drift.
- 7) Modelo baseline interpretable: logistic regression calibrada; modelo challenger: gradient-boosted trees. Validar con split temporal out-of-time.
- 8) Métricas: PR-AUC, ROC-AUC, lift/top-decile, recall a capacidad operativa, precision, calibration/Brier y estabilidad temporal.
- 9) Solo después de calibración convertir probabilidad a un índice 0-100 con bandas operativas. El nombre recomendado por ahora es Alert Propensity / Vulnerability Index hasta validar la interpretación de negocio.
- 10) Gobernanza: excluir features sensibles/proxies no aprobados, documentar razón de cada feature y evaluar disparidades del score.

Definición de modelado recomendada

Unidad = party_id + observation_date. Lookback = 12 meses, con recent 3m y medium 6m. Label candidato inicial = primera alerta en próximos 3 meses, comparado contra horizontes 1m/6m/12m. El modelo debe aprender cambios respecto al baseline individual, no solo niveles absolutos.

11. Data products creados / utilizados

- bu_frd.party_alert_tenure_sample_censored
- bu_frd.trades_eda_2025_sample
- bu_frd.trades_account_activity_2025
- bu_frd.trades_account_profile_2025
- bu_frd.trades_mop_transition_2025_v2
- bu_frd.trades_behavior_sample_2024_2025
- bu_frd.trades_mop_transition_2024_2025
- bu_frd.party_trade_behavior_monthly_2024_2025_v2
- bu_frd.party_trade_behavior_rolling_2025
- bu_frd.party_trade_features_2025_stage1
- bu_frd.party_trade_features_2025_stage2

English Version

Property	Value
Status	In progress - EDA and feature engineering
Objective	Build a point-in-time index/model for propensity to place a first fraud alert.
Primary sources	security_alert, person_restricted, trades
Analysis period	Source history 2000-2026; detailed behavioral EDA 2024-2025
Last updated	24-Aug-2026

1. Executive summary

- The alert source is a historical snapshot table. Creation events were therefore deduplicated and the earliest observed alert per party_id is used as the initial outcome.
- The tenure analysis found material left censoring in person_restricted: 29,086 of 66,643 sampled parties (43.64%) appear exactly on the first available historical date, 2000-09-30.
- After excluding censored and inconsistent records, 37,492 parties had usable tenure. Median tenure at first alert was 195 months (~16.3 years), but the peak shifts by +12 months each calendar year, indicating a cohort effect rather than a causal tenure threshold.
- trades is loaded monthly and supports 3m/6m/12m behavioral features. data_date is treated as information availability/load date and eff_dte as the account effective/last-verification date.
- Effective-update frequency is strongly explained by account lifecycle: 96.61% of STATIC accounts are closed; among accounts with no closed date, 86.06% are MONTHLY.
- MOP transitions provide a clean behavioral signal: 99.98% of deterioration transitions coincide with a new eff_dte. At the party level, 15.15% of the 2025 sample experienced at least one deterioration.
- The 2024-2025 monthly layer contains 707,157 party-months, 31,189 parties, 24 months, and zero party_id + data_date duplicates. 96.58% of 2025 party-months have a complete 12-month baseline.
- Among complete-baseline observations, 8.09% show recent deterioration, 6.12% show positive deterioration acceleration versus their own baseline, and 2.10% show a recent active-adverse event.

Primary methodological decision

Tenure should remain a contextual/segmentation feature. The primary index candidates are recent behavioral changes relative to each party's own baseline: MOP deterioration, past due, balance/utilization, adverse events, and portfolio changes.

2. Alert source and outcome definition

Metric	Result
security_alert source rows	324,268,295
Unique creation events	42,318,245
Parties with alert	29,766,492
Min / max crat_dte	2007-01-13 / 2026-07-30
Min / max data_date	2009-03-31 / 2026-07-31
First-alert cohort 2022-2025	6,663,521 parties
Type 2 in cohort	6,538,126 (98.12%)
Type 3 in cohort	122,420 (1.84%)
Types 2 and 3 same date	2,975 (0.04%)

- The business rule states that after placing an alert, a person cannot place another alert for approximately seven years.
- Many creation dates occur within seven years, so a different crat_dte cannot automatically be interpreted as a new independent alert. For the current objective, the earliest observed date is used as the first alert.
- The current outcome is propensity to place a first observed alert; it is not equivalent to confirmed fraud or confirmed victimization.

3. Tenure findings and revised interpretation

Indicator	Result
Deterministic alert sample	66,643 parties
Left-censored at 2000-09-30	29,086 (43.64%)
Uncensored	37,557

Valid after quality filters	37,492
Average tenure	175.23 months (~14.6 years)
P10 / P25	61 / 121 months
P50	195 months (~16.3 years)
P75 / P90 / P95	230 / 257 / 270 months

Alert year	Peak tenure band	% within year
2022	180-191 months	12.39%
2023	192-203 months	11.93%
2024	204-215 months	11.74%
2025	216-227 months	10.20%

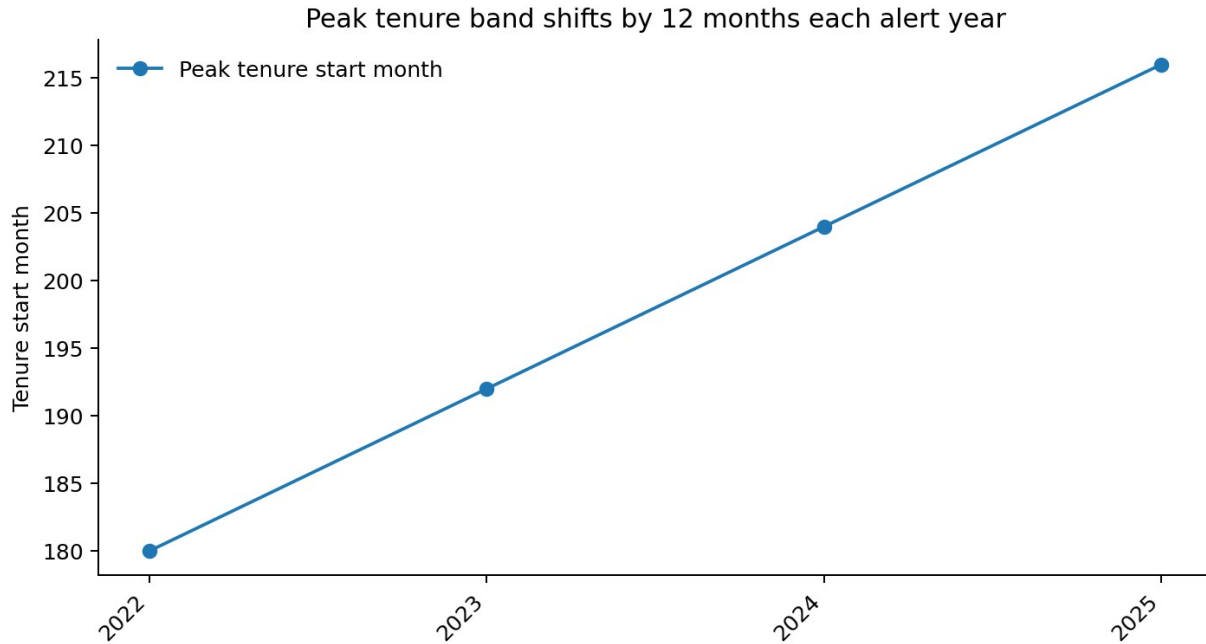


Figure 1. The peak tenure band shifts by exactly 12 months per year, consistent with a cohort effect.

Interpretation

The data do not support concluding that 17-19 years is the highest-risk tenure. The annual shift indicates that a large cohort (approximately originating in 2006-2007) is aging with the calendar. Estimating actual tenure risk requires the exposed-person denominator, including parties that do not place alerts.

4. trades structural and temporal EDA

Metric	Result
trades rows	504,842,538,762
Load frequency	Monthly
Observed history	From 2000-09 through 2026
Missing eff_dte	0
eff_dte > data_date	20 rows; effectively 0%
Key validated in sample	party_id + acct_id + data_date; 0 duplicates

Percentile	DATEDIFF(data_date, eff_dte)
P10	12 days
P25	25 days
P50	275 days
P75	1,671 days
P90	2,615 days
P95	3,073 days
P99	3,501 days

Correct interpretation of eff_dte

data_date - eff_dte measures information staleness/freshness, not simply load lag. A monthly load can carry forward the same effective account state for months or years.

5. Account lifecycle and update frequency

Update behavior	Accounts	%
STATIC (1 eff_dte)	167,962	51.50%
INTERMEDIATE (2-9)	29,323	8.99%
MONTHLY (10-12)	128,834	39.51%

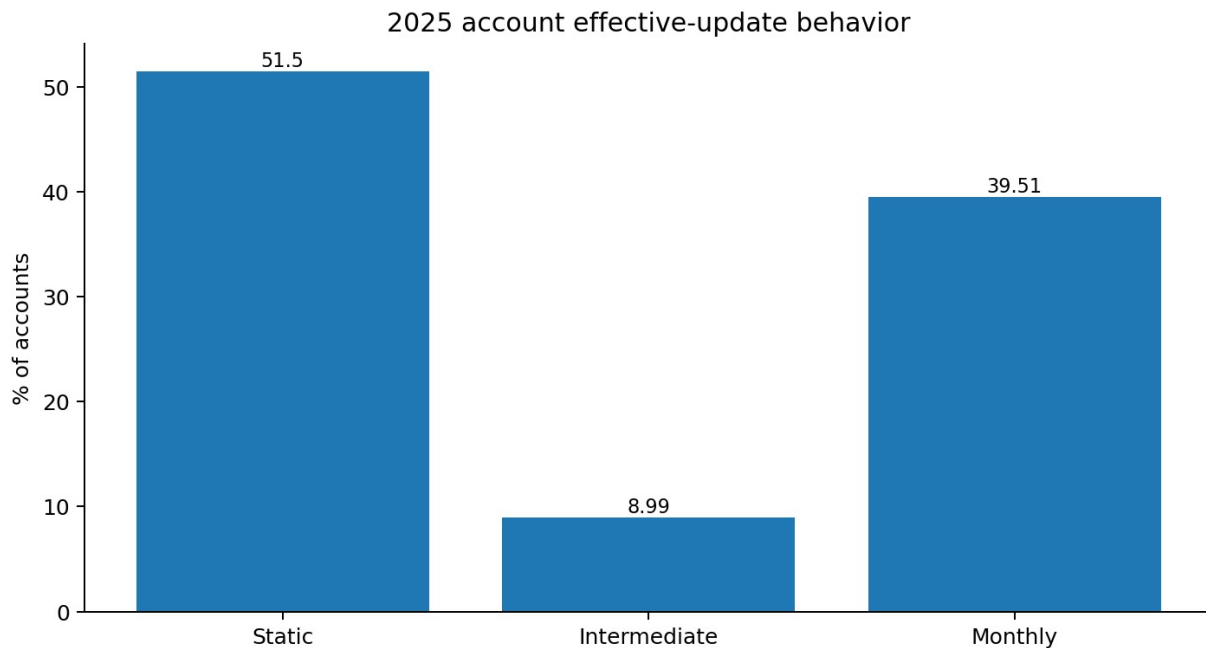


Figure 2. Effective-update frequency distribution in the 2025 sample.

Group	Closed	Zero balance	Past due
STATIC	96.61%	99.06%	0.69%
INTERMEDIATE	55.87%	61.38%	7.79%
MONTHLY	10.76%	26.31%	8.63%

Group (no clsd_dte only)	Accounts	%
STATIC	5,689	4.26%
INTERMEDIATE	12,940	9.69%
MONTHLY	114,968	86.06%

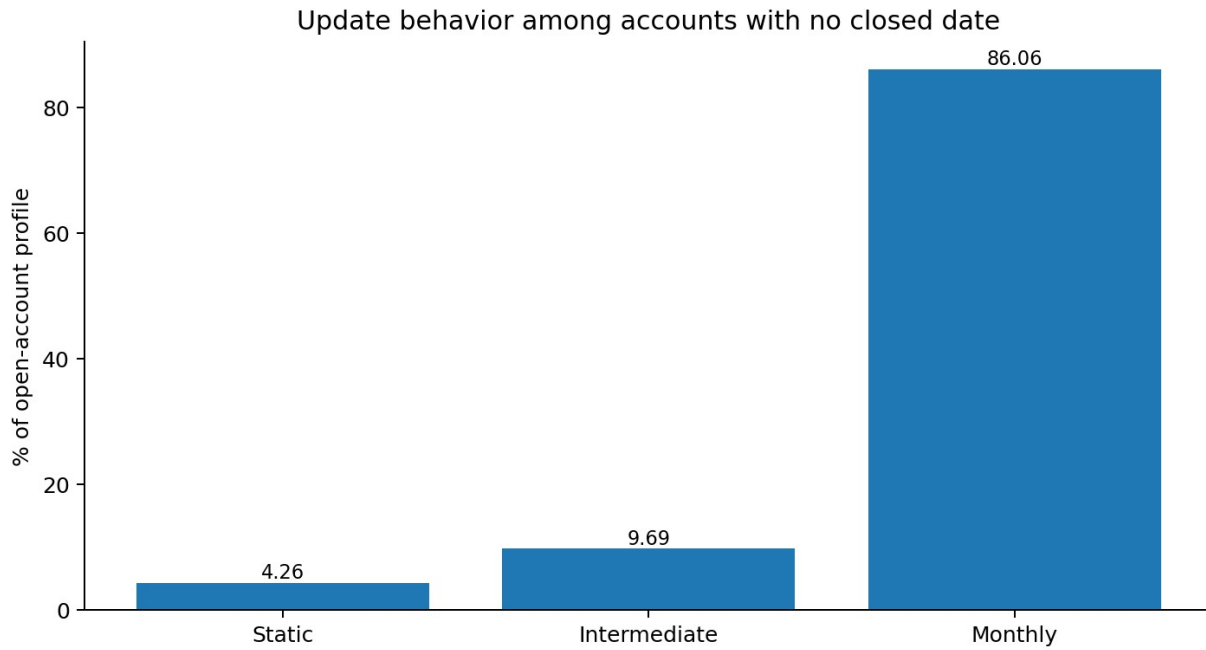


Figure 3. After controlling for accounts with no closed date, monthly updating is the dominant pattern.

Conclusion

Raw update_behavior mainly reflects lifecycle/reporting rather than direct vulnerability. It is retained as context; the higher-value signal is change relative to the account/party historical pattern.

6. First behavioral EDA: MOP transitions

Transition type	Transition count	%
NO_CHANGE	3,501,681	98.79%
DETERIORATION	29,310	0.83%
IMPROVEMENT	7,843	0.22%
NEW_ADVERSE_STATUS (v1)	4,042	0.11%
OTHER_TRANSITION	1,582	0.04%

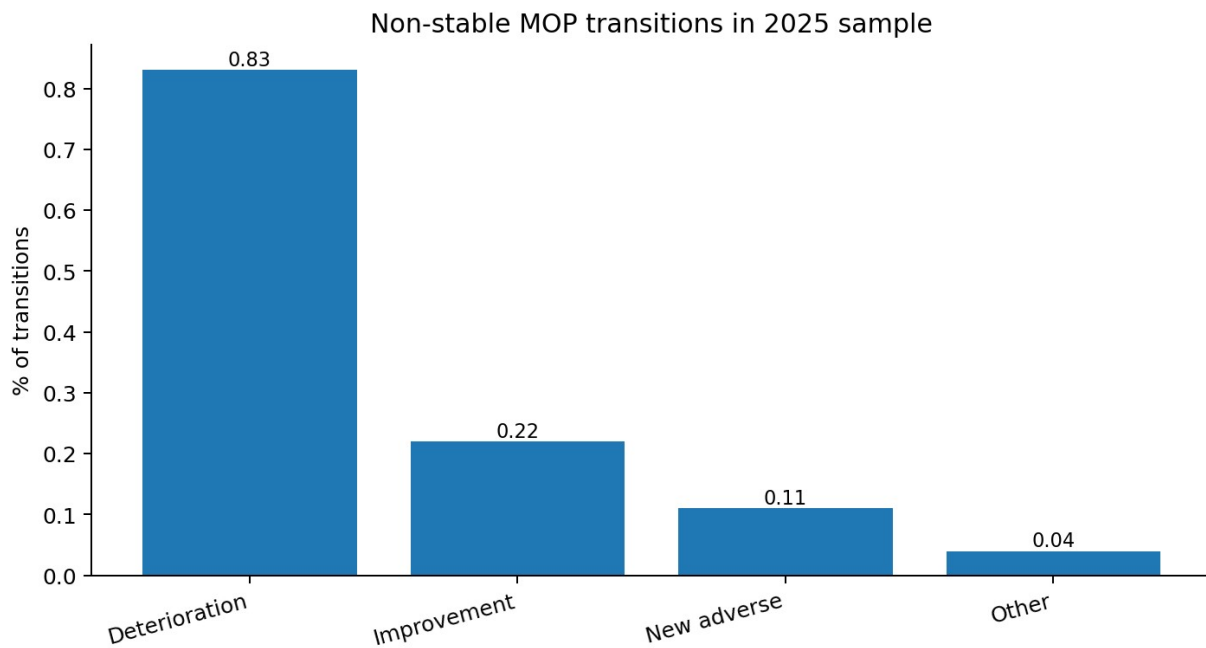


Figure 4. Non-stable transitions are rare at account-month level but behaviorally informative.

Temporal validation	% with effective_update_flag=1
Deterioration	99.98%
Improvement	99.78%
New adverse status	99.01%

- The ordinal 01-06 logic was validated for deterioration/improvement. Common observed changes include 01->02, 04->05, 02->03, 01->04, and 03->04.
- Adverse logic was corrected: 09, 9B, 8A, and 8B are active-adverse states; 9P and 8P are separated as historical/resolved states.
- Codes 07 and 08 remain uninterpreted pending an official definition.

Corrected MOP event	Count
DPD deterioration	29,310
DPD improvement	7,843
Entered active adverse	3,394
Adverse resolved	477
Became unrated	1,136
Left unrated	107

7. Party-month aggregation and 2024-2025 extension

Metric	Result
2025 parties in MOP EDA	30,496
Parties with at least one deterioration	4,624 (15.15%)
Parties with active adverse	1,697 (~5.6%)
P90 / P95 / P99 deteriorated accounts per party	2 / 6 / 19

2024-2025 monthly layer	Result
party-month rows	707,157
parties	31,189
months	24
range	2024-01-31 to 2025-12-31
party_id + data_date duplicates	0

2025 events with 2024 history available	Count
Deteriorations	31,437
Improvements	8,331
Active adverse	3,893

Boundary effect corrected

With 2024 history available, January 2025 can be compared with December 2024. The increase in 2025 detected events is expected and reflects improved longitudinal coverage rather than an inconsistency.

8. Rolling windows and behavioral baseline

2025 coverage	Party-months	%
Complete 12m baseline	343,948	96.58%
Incomplete	12,173	3.42%

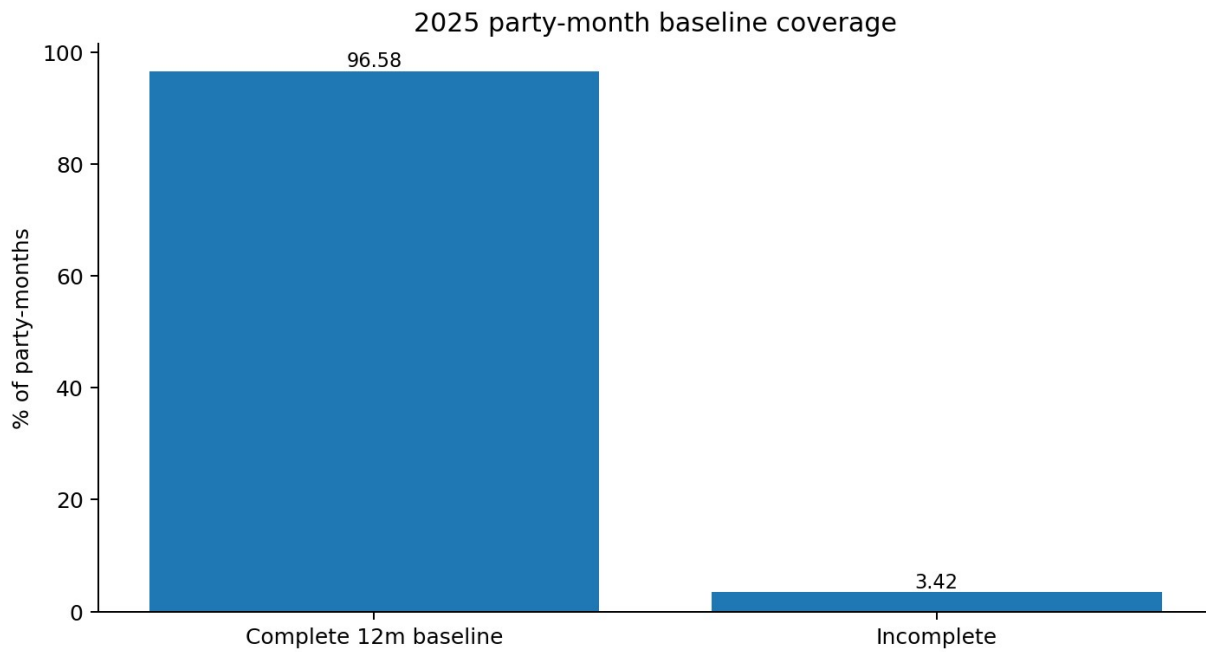


Figure 5. Complete-history coverage is sufficient for recent-versus-baseline comparisons.

Behavioral signal	Party-months	%
Deterioration in recent 3m	27,838	8.09%
Deterioration acceleration > 0	21,053	6.12%
Active adverse in recent 3m	7,216	2.10%

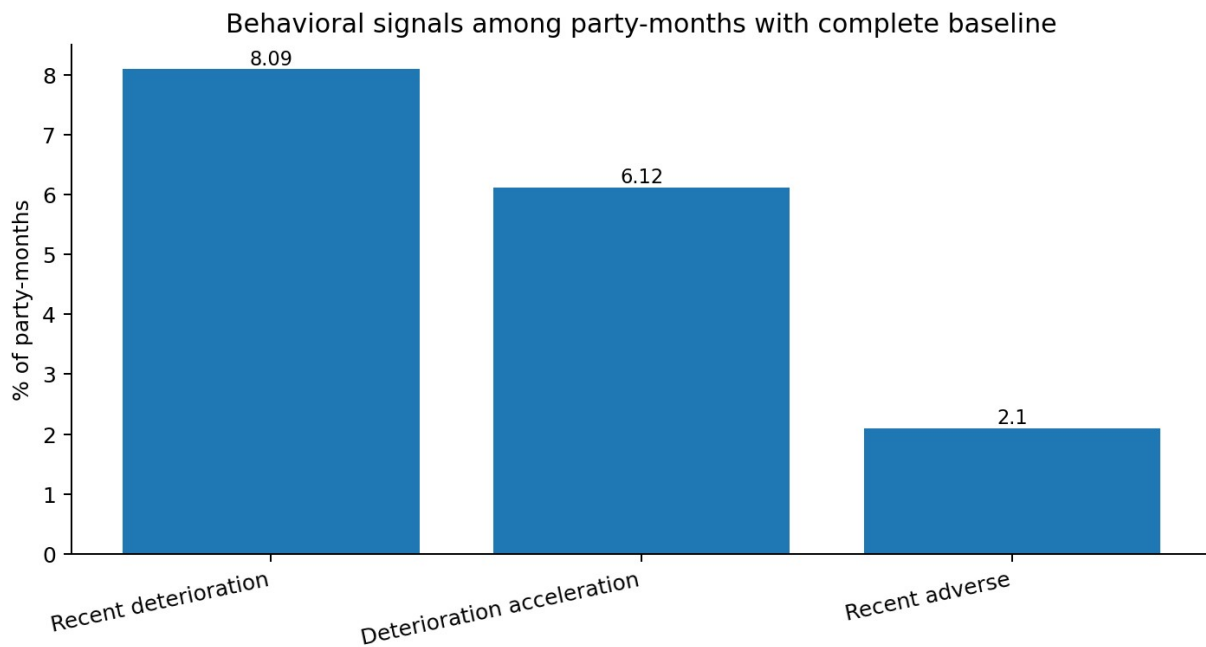


Figure 6. Frequency of the first behavioral features among complete-baseline observations.

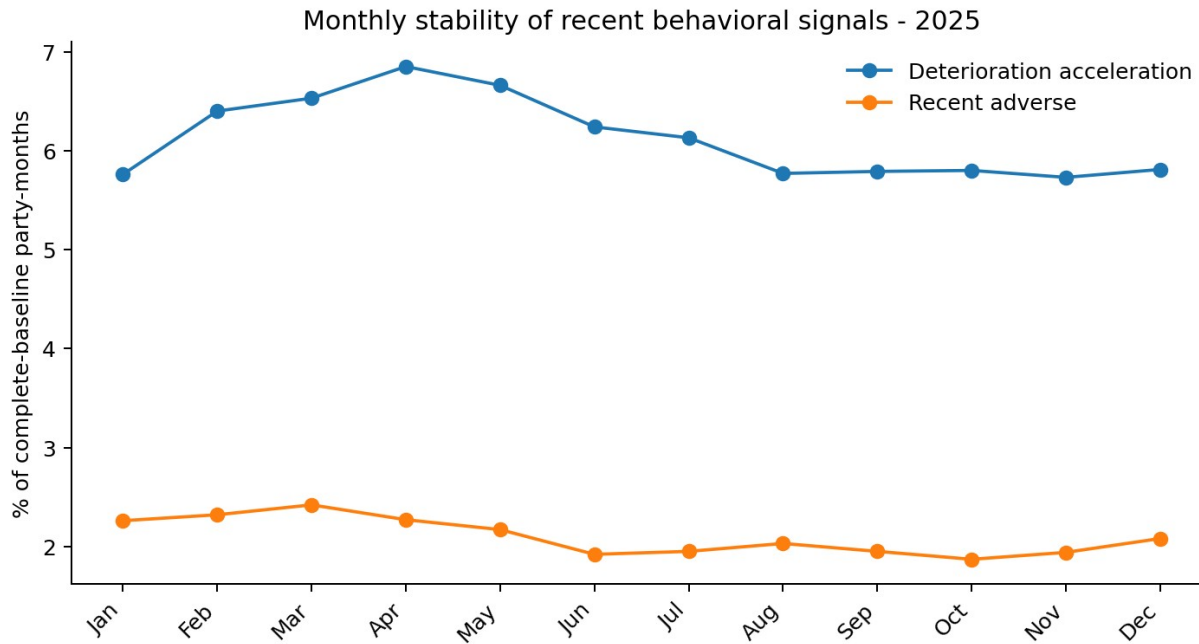


Figure 7. Deterioration acceleration and recent-adverse rates show reasonable monthly stability during 2025.

- pct_deterioration_acceleration ranges approximately from 5.73% to 6.85% by month in 2025.
- pct_recent_adverse ranges approximately from 1.87% to 2.42%.
- The P95 of deterioration_acceleration remains around 0.2-0.3 over the year, suggesting reasonable calendar stability for this feature.
- The pasted percentile output for past_due/balance/utilization change magnitudes was malformed; it is intentionally excluded pending a clean re-export.

9. Feature Catalog - current status

Feature family / feature	Priority	Current decision
tenure_months	Medium	Context/segmentation; not isolated risk
active_accounts	High	Exposure control / denominator
deteriorations_3m/6m/12m	Very high	Primary candidate
deterioration_acceleration	Very high	Recent 3m vs prior 9m change
max_dpd_jump_3m	High	Observed deterioration severity
entered_active_adverse_3m/6m	Very high	Recent severe event
past_due_level/change	Very high	Level and change vs baseline
revolving_utilization/change	Very high	R/C portfolios only; validate outliers
new / recently closed accounts	High	Feature engineering pending
raw_update_behavior	Low	Lifecycle/reporting, not direct risk
staleness / update-frequency change	Experimental	Use only relative to baseline
legis_prhib / medical classifications	Exclude	Governance/sensitivity; not score
DOB/age/state	Audit	Not default first-version production features

10. Next steps to build the index

- 1) Cleanly re-export and validate percentiles for past_due_change, balance_change, utilization_change, and delinquency_share_change.
- 2) Build the point-in-time observation table. For each alert, use the latest data_date strictly before first_alert_date; never use a load after the event.
- 3) Build alert_next_1m, alert_next_3m, alert_next_6m, and alert_next_12m labels. 1 = first alert inside the future window; 0 = no alert with complete follow-up; NULL = censored.
- 4) Build the eligible no-alert/control universe and denominator. This is required to move from descriptive concentration to actual alert risk/rate.
- 5) Align 3m/6m/12m and recent-3m-vs-prior-9m features to the observation date with strict point-in-time controls and no leakage.

- 6) Run label-based EDA: distributions, lift, event rates by bands, missingness, monotonicity, month/year stability, and drift.
- 7) Interpretable baseline model: calibrated logistic regression; challenger model: gradient-boosted trees. Validate with an out-of-time split.
- 8) Metrics: PR-AUC, ROC-AUC, lift/top-decile, recall at operational capacity, precision, calibration/Brier, and temporal stability.
- 9) Only after calibration, convert probability into a 0-100 operational index with risk bands. Recommended working name: Alert Propensity / Vulnerability Index until business interpretation is validated.
- 10) Governance: exclude unapproved sensitive/proxy features, document the rationale for each feature, and test score disparities.

Recommended modeling definition

Unit = party_id + observation_date. Lookback = 12 months, with a recent 3-month window and a medium 6-month window. Initial candidate label = first alert in the next 3 months, benchmarked against 1m/6m/12m horizons. The model should learn changes relative to the individual baseline rather than only absolute levels.

11. Data products created / used

- bu_frd.party_alert_tenure_sample_censored
- bu_frd.trades_eda_2025_sample
- bu_frd.trades_account_activity_2025
- bu_frd.trades_account_profile_2025
- bu_frd.trades_mop_transition_2025_v2
- bu_frd.trades_behavior_sample_2024_2025
- bu_frd.trades_mop_transition_2024_2025
- bu_frd.party_trade_behavior_monthly_2024_2025_v2
- bu_frd.party_trade_behavior_rolling_2025
- bu_frd.party_trade_features_2025_stage1
- bu_frd.party_trade_features_2025_stage2

End of bilingual Confluence-ready analytical progress report

Actualizacion analitica: labels point-in-time y primer test case-control

- Se construyeron labels point-in-time para la primera alerta a 1, 3, 6 y 12 meses. La fecha maxima observable de creacion de alerta es 2026-07-30 y la fecha maxima de snapshot es 2026-07-31.
- El dataset elegible para labels contiene 312,754 party-months. Los horizontes 1m, 3m y 6m tienen follow-up completo en 2025; el horizonte 12m presenta 156,417 observaciones censuradas, aproximadamente la mitad del universo disponible.
- El horizonte de 3 meses se mantiene como candidato principal por accionabilidad y cobertura; 6 meses se conserva como candidato secundario. La muestra general contiene 149 parties positivos unicos a 3 meses y 176 a 6 meses.

Horizonte	Party-months positivos	Negativos	Censurados	Parties positivos unicos
1 mes	116	312,638	0	116
3 meses	369	312,385	0	149
6 meses	741	312,013	0	176
12 meses	1,294	155,043	156,417	184

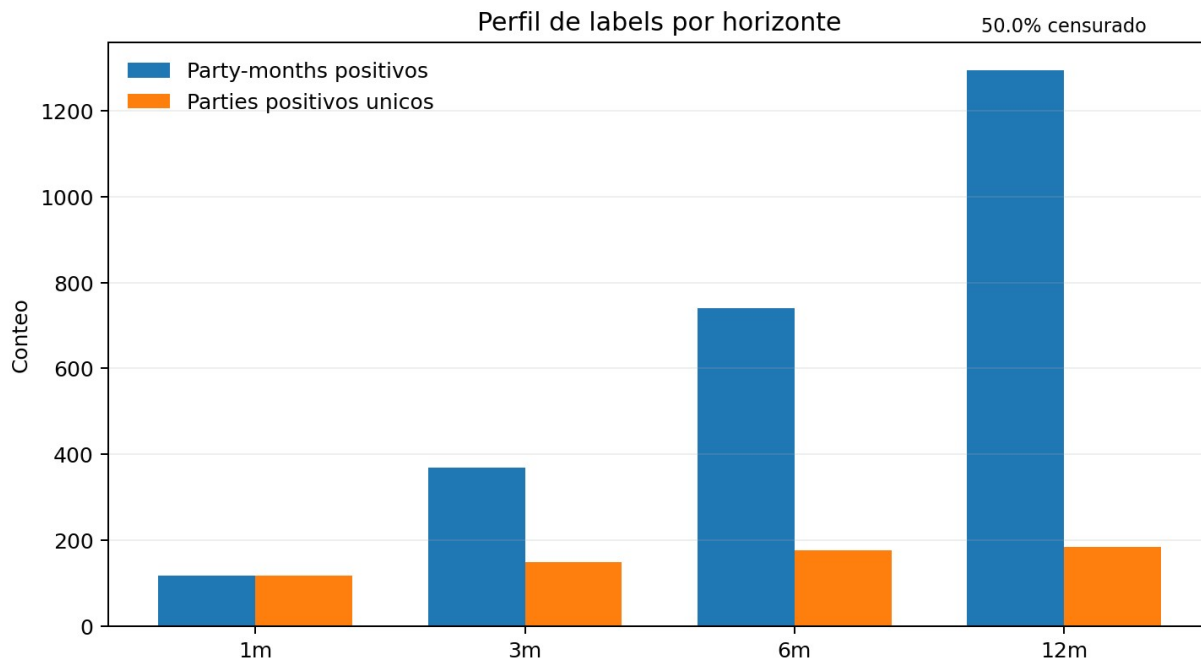


Figura 8. Perfil de labels por horizonte. El horizonte de 12 meses pierde aproximadamente 50% de las observaciones por censura derecha.

Diseno del primer EDA case-control

- Para evitar contar varias veces el mismo evento, se creo una cohorte positiva con una sola observacion por party_id: la primera observacion mensual que entra en la ventana positiva de 3 meses antes de la primera alerta. El resultado fue 149 casos unicos distribuidos de enero a diciembre de 2025.
- Los controles se seleccionaron 20:1 dentro del mismo data_date y active_accounts_band. El estrato con menor disponibilidad tenia aproximadamente 398 controles disponibles por caso, por lo que no hubo restriccion material de matching.
- El dataset final contiene 149 casos y 2,980 observaciones control, correspondientes a 2,784 parties control unicos. Algunas personas control aparecen en mas de un mes; esto es aceptable para EDA, pero el modelado debera agrupar por party_id y respetar el tiempo.

Metrica	Casos	Controles
Observaciones	149	2,980
Parties unicos	149	2,784
Promedio active_accounts	7.22	7.21
Mediana active_accounts	6	6
P90 active_accounts	14	15

- Las bandas de cuentas activas quedaron balanceadas por construccion: 1.34% con cero cuentas activas, 6.71% con una, 17.45% con 2-3, 23.49% con 4-5, 28.19% con 6-10 y 22.82% con 11 o mas.

Primera evidencia de senal

Feature	Casos	Controles	Prevalencia relativa	Lectura preliminar
Deterioro reciente 3m	10.07%	9.66%	1.04x	Separacion minima
Aceleracion del deterioro	10.07%	7.25%	1.39x	Senal direccional prometedora
Adverse reciente	2.68%	1.78%	1.51x	Prometedor, pero solo 4 casos

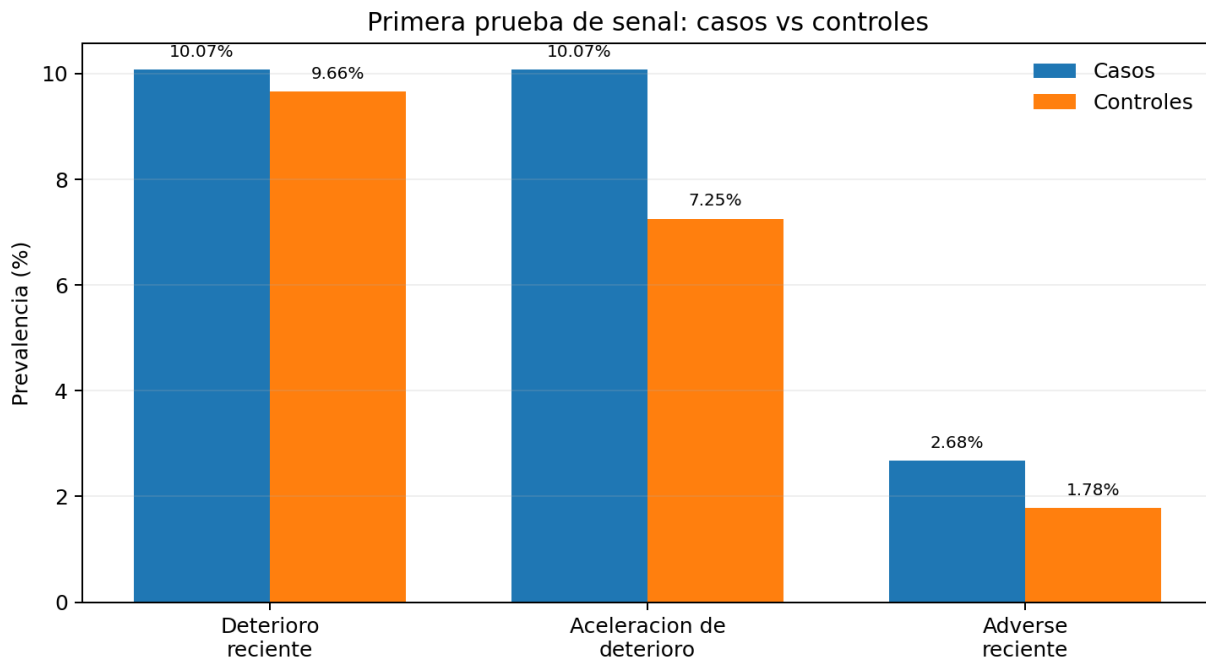


Figura 9. Primera comparacion case-control. La aceleracion del deterioro respecto al baseline individual muestra una separacion mayor que el deterioro reciente aislado.

Interpretacion actual

El resultado mas relevante es que el deterioro reciente aislado casi no separa casos de controles (10.07% vs 9.66%), mientras que la aceleracion del deterioro frente al baseline personal aparece con mayor frecuencia en casos (10.07% vs 7.25%; 1.39x). Esto respalda la estrategia de modelar cambios relativos al comportamiento historico individual. Recent adverse tambien muestra una diferencia direccional (1.51x), pero solo 4 de 149 casos presentan el evento, por lo que la evidencia aun es escasa.

- Estos resultados son exploratorios, no concluyentes: 149 casos es una muestra pequena para definir pesos de un score o afirmar significancia estadistica robusta.
- Recent deterioration pasa a prioridad media-baja como feature aislada. Deterioration acceleration pasa a prioridad alta experimental. Recent adverse permanece como feature experimental de alta prioridad, condicionada a ampliar la muestra positiva.

Siguientes pasos acordados

Paso	Objetivo	Accion
1	Comparar magnitud	Comparar percentiles de deterioration acceleration, recent deterioration count, max DPD jump, past due, delinquency share, utilization y balance entre casos y controles.
2	Evaluar convergencia	Medir si varias senales cambian simultaneamente antes de la alerta, no solo de forma aislada.
3	Repetir a 6 meses	Usar 6m como horizonte secundario y comparar direccion/estabilidad de la senal.
4	Enriquecer positivos	Incluir todos los casos elegibles de alerta y una muestra representativa de controles para desarrollo de modelo.
5	Validacion out-of-time	Separar train/validation/test por tiempo y mantener todas las observaciones de cada party_id en un solo split.
6	Indice/modelo	Entrenar y calibrar modelos, evaluar PR-AUC, lift, recall a capacidad, calibracion, estabilidad temporal y fairness antes de definir el score productivo.

Analytical update: point-in-time labels and first matched case-control signal test

- Point-in-time labels were created for the first observed alert at 1, 3, 6 and 12 months. The maximum observable alert creation date is 2026-07-30 and the maximum snapshot date is 2026-07-31.
- The label-eligible dataset contains 312,754 party-months. The 1m, 3m and 6m horizons have complete follow-up for the 2025 observations; the 12m horizon has 156,417 censored observations, roughly half of the available universe.
- The 3-month horizon remains the primary candidate because it balances actionability and coverage; 6 months remains the secondary candidate. The general sample contains 149 unique positive parties at 3 months and 176 at 6 months.

Horizon	Positive party-months	Negatives	Censored	Unique positive parties
1 month	116	312,638	0	116
3 months	369	312,385	0	149
6 months	741	312,013	0	176
12 months	1,294	155,043	156,417	184

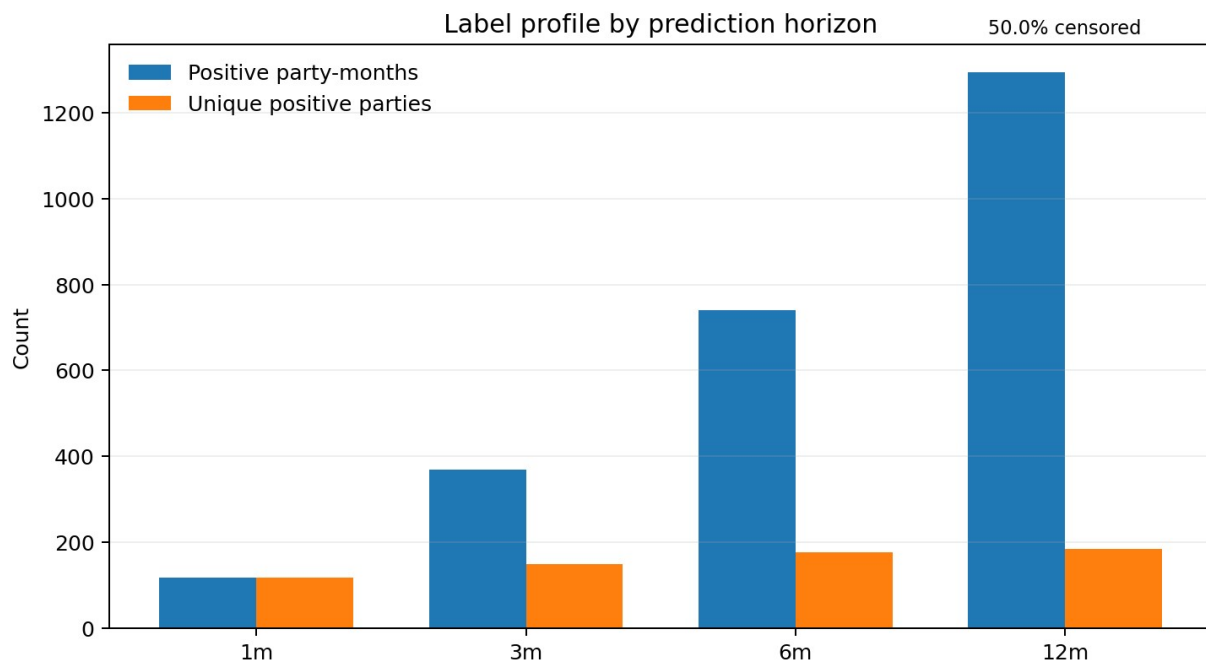


Figure 8. Label profile by prediction horizon. The 12-month horizon loses approximately 50% of observations to right censoring.

First matched case-control EDA design

- To avoid counting the same event multiple times, the positive cohort uses one observation per party_id: the earliest monthly observation entering the positive 3-month window before the first alert. This produced 149 unique cases distributed from January through December 2025.
- Controls were selected 20:1 within the same data_date and active_accounts_band. The most restrictive stratum still had about 398 available controls per case, so matching availability was not a material limitation.
- The final dataset contains 149 cases and 2,980 control observations representing 2,784 unique control parties. Some control parties appear in more than one month; this is acceptable for EDA, but modeling must group by party_id and preserve temporal separation.

Metric	Cases	Controls
Observations	149	2,980
Unique parties	149	2,784
Mean active_accounts	7.22	7.21
Median active_accounts	6	6
P90 active_accounts	14	15

- Active-account bands are balanced by construction: 1.34% with zero active accounts, 6.71% with one, 17.45% with 2-3, 23.49% with 4-5, 28.19% with 6-10, and 22.82% with 11 or more.

First evidence of signal

Feature	Cases	Controls	Relative prevalence	Preliminary read
Recent deterioration 3m	10.07%	9.66%	1.04x	Minimal separation
Deterioration acceleration	10.07%	7.25%	1.39x	Promising directional signal
Recent adverse	2.68%	1.78%	1.51x	Promising, but only 4 cases

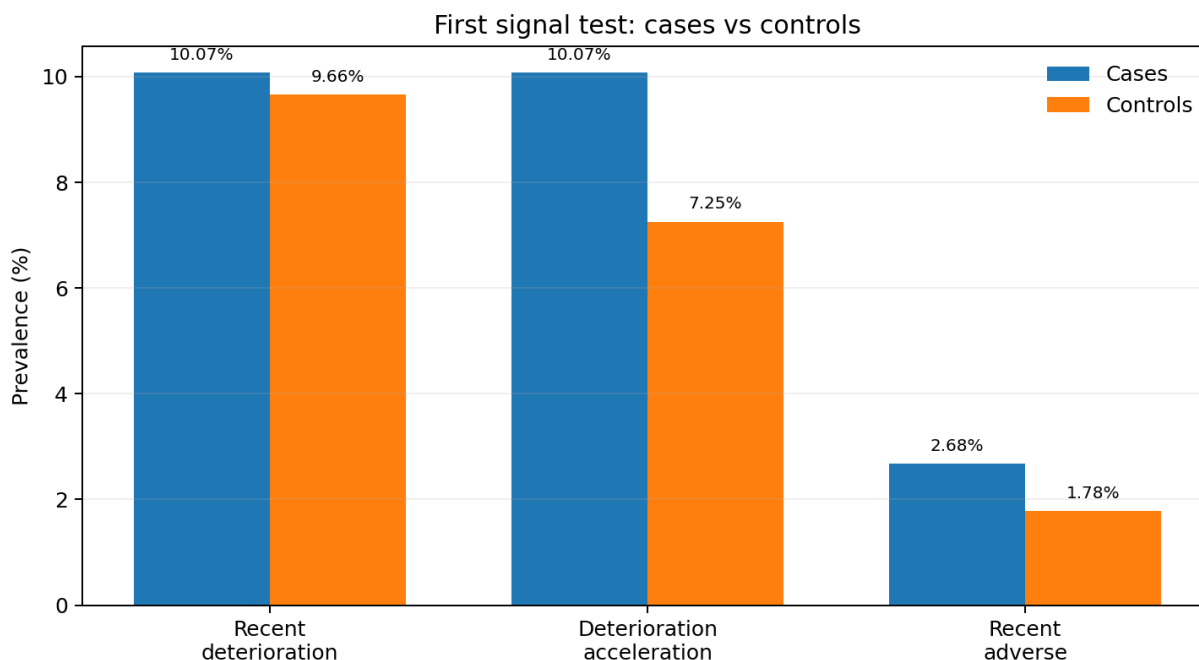


Figure 9. First matched case-control comparison. Deterioration acceleration relative to each party's own baseline separates the groups more than raw recent deterioration.

Current interpretation

The most relevant result is that raw recent deterioration barely separates cases from controls (10.07% vs 9.66%), while deterioration acceleration relative to the party's own baseline is more common in cases (10.07% vs 7.25%; 1.39x). This supports modeling behavioral change relative to individual history. Recent adverse is also directionally higher (1.51x), but only 4 of 149 cases have the event, so evidence remains sparse.

- These findings are exploratory rather than conclusive: 149 cases are too few to define production score weights or establish robust statistical significance.
- Recent deterioration is downgraded to medium-low priority as a standalone feature. Deterioration acceleration is promoted to high experimental priority. Recent adverse remains a high-priority experimental feature, conditional on a larger positive sample.

Agreed next steps

Step	Objective	Action
1	Compare magnitude	Compare percentiles of deterioration acceleration, recent deterioration count, max DPD jump, past due, delinquency share, utilization and balance between cases and controls.
2	Evaluate convergence	Test whether multiple behavioral signals move together before the alert rather than only in isolation.
3	Repeat at 6 months	Use 6m as the secondary horizon and compare signal direction and stability.
4	Enrich positives	Include all eligible alert-positive cases plus a representative control sample for model development.
5	Out-of-time validation	Split train/validation/test by time and keep all observations for each party_id in one split.
6	Index/model	Train and calibrate models; evaluate PR-AUC, lift, recall at capacity, calibration, temporal stability and fairness before defining a production score.